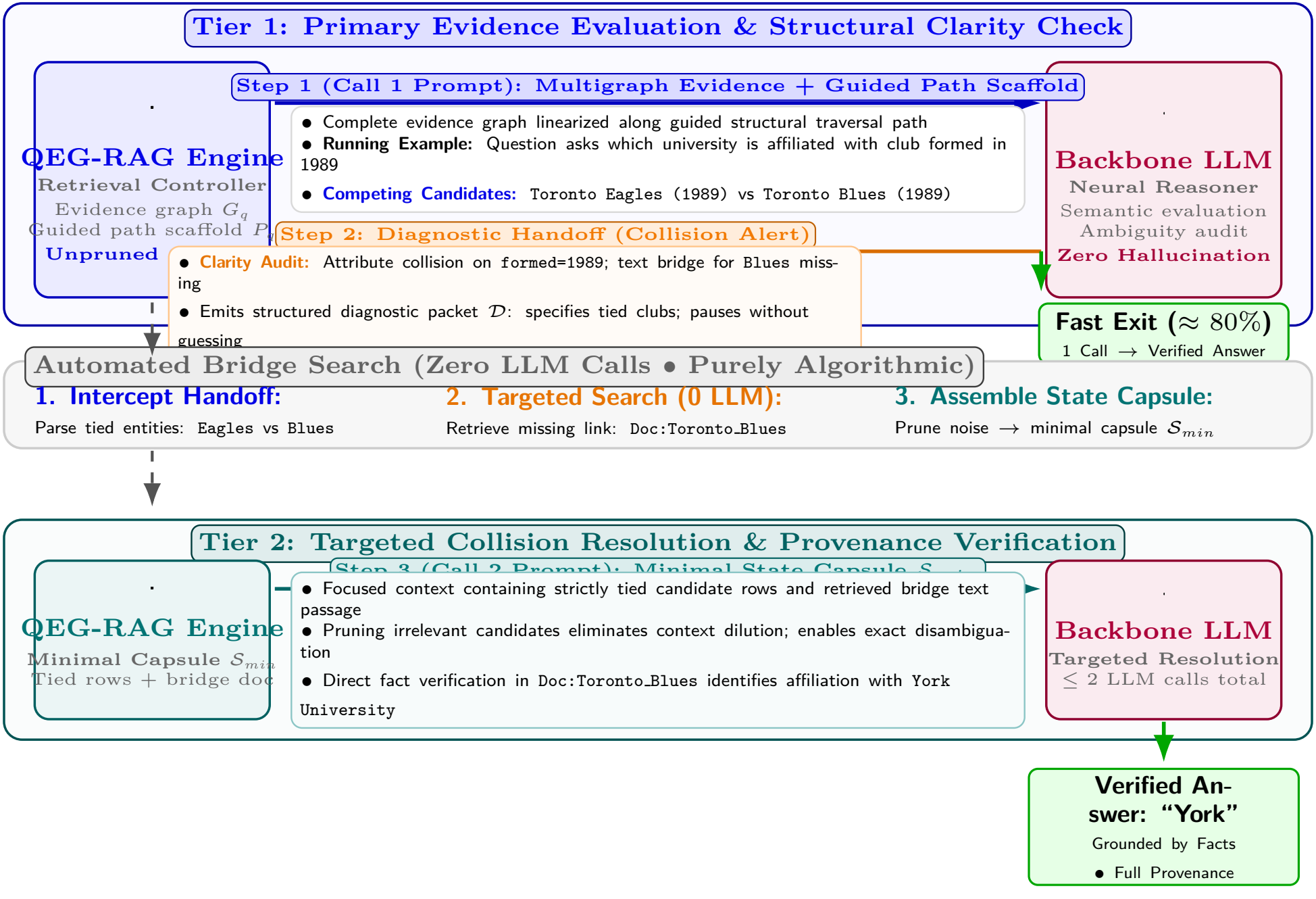

Tier 1: Primary Evidence Evaluation & Structural Clarity Check
QEG-RAG Engine
Retrieval Controller
Evidence graph G_q
Guided path scaffold P_q
Unpruned
Step 1 (Call 1 Prompt): Multigraph Evidence + Guided Path Scaffold
• Complete evidence graph linearized along guided structural traversal path
• Running Example: Question asks which university is affiliated with club formed in 1989
• Competing Candidates: Toronto Eagles (1989) vs Toronto Blues (1989)
Backbone LLM
Neural Reasoner
Semantic evaluation
Ambiguity audit
Zero Hallucination
Step 2: Diagnostic Handoff (Collision Alert)
• Clarity Audit: Attribute collision on formed=1989; text bridge for Blues missing
• Emits structured diagnostic packet $\mathcal{D}$: specifies tied clubs; pauses without guessing
Fast Exit (≈ 80%)
1 Call → Verified Answer
Automated Bridge Search (Zero LLM Calls • Purely Algorithmic)
1. Intercept Handoff:
Parse tied entities: Eagles vs Blues
2. Targeted Search (0 LLM):
Retrieve missing link: Doc:Toronto_Blues
3. Assemble State Capsule:
Prune noise → minimal capsule $\mathcal{S}_{min}$
Tier 2: Targeted Collision Resolution & Provenance Verification
QEG-RAG Engine
Minimal Capsule $\mathcal{S}_{min}$
Tied rows + bridge doc
• Focused context containing strictly tied candidate rows and retrieved bridge text passage
• Pruning irrelevant candidates eliminates context dilution; enables exact disambiguation
• Direct fact verification in Doc:Toronto_Blues identifies affiliation with York University
Backbone LLM
Targeted Resolution
≤ 2 LLM calls total
Verified Answer: "York"
Grounded by Facts
• Full Provenance